

Beyond Citation Entailment: Causal Evidence Audits for Retrieval-Augmented Language Models

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Abstract

Retrieval-augmented language models and deep-research agents are commonly evaluated by answer correctness and by whether cited passages support generated claims. These observations establish compatibility between an answer and its citations, but they do not establish that the cited evidence determined the answer. A system can answer from parametric memory, a positional shortcut, or a prior draft and attach an entailing citation afterward. We introduce the *Causal Evidence Audit* (CEA), a black-box evaluation protocol that complements citation entailment with three controlled tests: paired-world responsiveness when one decisive fact changes, abstention when indispensable evidence is removed, and proof-chain citation coverage. We show constructively that causal grounding is not identifiable from any single-world observational citation score. We then instantiate the protocol in CEVAL-32, a transparent pilot of 32 fictional-entity items spanning direct lookup, two-hop composition, numerical comparison, and conditional reasoning. Two locally executed open-weight models, each tested with ordinary and evidence-contract prompts, produced 384 greedy-decoded generations. Single-world observational pass rates ranged from 59.4% to 78.1%, whereas the coverage-based joint causal score ranged from 0% to 31.3%; the strict score, which also rejects extraneous citations, ranged from 0% to 31.3%. Stronger grounding instructions sometimes increased the observational score while decreasing the strict causal score. These pilot results are not a model leaderboard. They demonstrate that commonly reported citation success can conceal unsupported guessing and post-hoc attribution, and motivate paired evidence interventions as a minimum evaluation layer for high-stakes RAG and research agents.

1 Introduction

Retrieval-augmented generation (RAG) was introduced to combine parametric language modeling with external, updateable knowledge [Lewis et al., 2020]. The basic promise is operationally important: retrieve relevant evidence, generate an answer conditioned on it, and expose sources that a reader can inspect. Browser-assisted systems such as WebGPT [Nakano et al., 2021] and evidence-producing models such as GopherCite [Menick et al., 2022] made citations a central interface for trustworthy

language generation. The same pattern now underlies long-form research agents, which browse, synthesize, and produce citation-rich reports.

Evaluation has largely followed the visible artifact. Attributable to Identified Sources (AIS) asks whether generated statements are supported by identified sources [Rashkin et al., 2021]. Audits of generative search measure citation completeness and citation correctness [Liu et al., 2023]; ALCE evaluates fluency, answer correctness, and citation quality [Gao et al., 2023]. RAGAS, ARES, and RAGCHECKER decompose retrieval relevance, faithfulness, and answer quality in increasingly fine-grained ways [Es et al., 2023, Saad-Falcon et al., 2023, Ru et al., 2024]. These are necessary measurements. They answer whether the output is consistent with the presented evidence and whether a reader can verify it.

They do not, by themselves, answer a different question: *did the system’s answer depend on the evidence it cites?* An entailing citation can be attached after an answer has already been produced. A model may ignore a decisive passage, follow a familiar prior, select the first document, or guess under missing evidence while still producing a citation that looks appropriate. Recent work makes the distinction increasingly consequential. ContextCite and RAGONITE use context removal to estimate which evidence contributed to a generation [Cohen-Wang et al., 2024, Saha Roy et al., 2024]. Counterfactual prompting and set-level verification have been used for RAG risk control and selective abstention [Chen et al., 2024, Qiu et al., 2026], while leave-one-out changes can reveal poisoned documents [Kumar et al., 2026]. Deceptive Grounding shows that real citations and high measured faithfulness can coexist with entity-attribution failure [Caruzzo et al., 2026]. Work on knowledge conflicts further shows that models may integrate evidence unevenly or positionally [Xu et al., 2024, Yang et al., 2026]. Meanwhile, evaluations of deep-research agents identify evidence integration and dependency-consistent reasoning as persistent bottlenecks [Zhang et al., 2025, Guo et al., 2026].

This paper argues that grounding evaluation needs both *observational support* and *interventional reliance*. We propose CEA, a model-agnostic protocol built around matched evidence worlds. In a paired item, exactly one answer-determining proposition is changed; nuisance content is either held fixed or modified by a predeclared, answer-invariant balancing edit. A grounded system

must change its answer accordingly. In a matched ablation, indispensable evidence is removed; a grounded system must abstain rather than complete the missing link from a shortcut. Citations are then checked against the oracle proof set in each world.

The contributions are:

1. a formal separation between semantic citation support and causal evidence reliance, including a constructive non-identifiability result for single-world observational metrics;
2. a black-box audit protocol and joint metrics that combine paired-world responsiveness, necessity-aware abstention, and proof-chain citation alignment;
3. CEVAL-32, a reproducible 32-item pilot with fictional entities and four reasoning families, together with 384 raw generations from two quantized open-weight models under two prompting regimes; and
4. an intervention-based training objective and reporting checklist for extending the protocol to real retrieval corpora and long-horizon research agents.

The empirical study is intentionally labeled a pilot. Its purpose is to test whether the proposed measurements reveal behavior hidden by a single-world score, not to rank model families or establish population-level performance.

2 Related Work

2.1 Attribution and RAG evaluation

AIS formalized whether a statement is attributable to identified sources and validated a human annotation procedure across generation tasks [Rashkin et al., 2021]. Generative-search audits subsequently operationalized citation recall and precision, finding substantial gaps despite fluent answers [Liu et al., 2023]. ALCE provided reproducible citation evaluation over several long-form question-answering settings [Gao et al., 2023]. RAGAS evaluates RAG pipelines without reference answers [Es et al., 2023]; ARES trains lightweight judges with synthetic data and uses prediction-powered inference [Saad-Falcon et al., 2023]; RAGCHECKER diagnoses retriever and generator failure modes at claim level [Ru et al., 2024]. These methods estimate support, relevance, utilization, or human preference from observed inputs and outputs.

Citation generation can occur with the answer or after it. A recent comparison reports a coverage-correctness trade-off between generation-time and post-hoc citation paradigms [Saxena et al., 2025]. That result makes reliance especially important: if a citation is selected after drafting, semantic support does not imply that the source influenced the draft. CEA is agnostic to when citations are produced and tests the completed system under controlled evidence changes.

2.2 Context attribution and interventions

ContextCite attributes a generated statement to parts of the context using a sparse surrogate trained on context ablations [Cohen-Wang et al., 2024]. RAGONITE similarly computes counterfactual attribution by comparing a response with the response obtained after removing evidence [Saha Roy et al., 2024]. Their central intuition—removing a causally relevant source should affect the response—is aligned with this work. Our focus is complementary: rather than estimate a contribution for an already generated statement, CEA defines an end-to-end evaluation contract across two coherent answer-changing worlds, a necessary-evidence ablation, oracle answers and proof sets, and the citations emitted in each world.

Counterfactual methods have also been used to estimate RAG risk and encourage abstention [Chen et al., 2024]. SURE-RAG treats evidence sufficiency as a set-level property and evaluates a selective verifier with counterfactual swaps and no-oracle controls [Qiu et al., 2026]. RAGuard uses leave-one-out decoding as one layer of a poisoning defense [Kumar et al., 2026]. Attribution-bias studies intervene on source authorship metadata and show that attribution changes under nominally irrelevant cues [Abolghasemi et al., 2024]. Deceptive Grounding isolates entity-attribution failures that remain invisible to conventional faithfulness and citation checks [Caruzzo et al., 2026]. These works establish that counterfactual removal, sufficiency verification, abstention, and attribution failure are not individually new. To our knowledge, prior protocols do not make correct behavior in both minimally different answer worlds, abstention after removing an indispensable proof element, and oracle proof-chain citations a single per-item pass condition. CEA targets that joint behavioral contract.

Counterfactual evaluation has a longer history in knowledge updating. MQuAKE asks whether edited facts propagate to multi-hop consequences [Zhong et al., 2023]; ReCoE broadens counterfactual editing to multiple reasoning schemes [Hua et al., 2024]. Those benchmarks evaluate changes to model knowledge. CEA instead intervenes on inference-time evidence and does not modify model weights.

2.3 Research agents and evidence integration

DeepResearch Bench assesses expert-level report tasks, effective citations, and citation accuracy [Du et al., 2025]. ReportBench checks cited-statement consistency and non-cited factuality in generated surveys [Li et al., 2025]. FINDER and its failure taxonomy identify evidence integration, verification, and resilient planning as core weaknesses [Zhang et al., 2025]. Mr.LHDR extends evaluation to long, multimodal dependency graphs and reports a large gap between final-answer accuracy and strict dependency-consistent success [Guo et al.,

2026]. These findings motivate an audit that can localize whether a research conclusion follows its decisive evidence. CEA is designed as an additional layer for such benchmarks, not as a replacement for coverage, writing quality, source authority, or factuality evaluation.

3 Why Citation Support Is Not Enough

3.1 Problem setting

Let q be a query and $E = (e_1, \dots, e_m)$ an ordered evidence bundle. A system A returns an answer and a citation set,

$$A(q, E) = (\hat{y}, C), \quad C \subseteq \{1, \dots, m\}. \quad (1)$$

Let $y^*(q, E)$ be the oracle answer, and let $G(q, E)$ be a minimal oracle proof set: the document indices required to derive y^* under the benchmark semantics. The abstention symbol $\perp$ denotes that E does not determine a unique answer.

An observational support evaluator sees one realized tuple $(q, E, \hat{y}, C)$. It may assess correctness, whether E_C entails $\hat{y}$, whether all answer claims are cited, and whether citations are relevant. These properties are functions of the realized artifact. Evidence reliance is a property of the mapping $E \mapsto A(q, E)$ across possible evidence worlds.

Definition 1 (Observational citation support). *For a support predicate S , an output is observationally supported when $S(q, E, \hat{y}, C) = 1$. The predicate may be implemented by humans, natural-language inference, an LLM judge, or a controlled proof oracle.*

Definition 2 (Interventional evidence responsiveness). *For a valid intervention T that changes answer-determining evidence, a system is responsive on (q, E, T) when its output changes to the oracle answer for $T(E)$ while remaining correct on E .*

3.2 A non-identifiability result

Proposition 1 (Single-world non-identifiability). *No evaluator that observes only $(q, E_0, \hat{y}, C)$ in one evidence world can, without additional assumptions, identify whether $\hat{y}$ causally depends on E_0 .*

Proof. Fix an observed world E_0 with oracle answer y_0 and an entailing citation set C_0 . Construct a responsive system A_R that computes $y^*(q, E)$ and cites its proof set. Construct a post-hoc system A_P that returns the memorized answer y_0 and attaches C_0 whenever evaluated at E_0 . The two systems induce the same observed tuple at E_0 , so every evaluator restricted to that tuple assigns the same score. Now choose a valid intervention T with $y^*(q, T(E_0)) = y_1 \neq y_0$. By construction, A_R returns

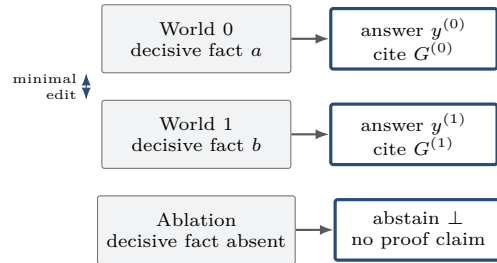


Figure 1: A three-condition Causal Evidence Audit. Passing requires correct behavior in all matched worlds, not merely an entailing citation in one world.

y_1 , while A_P can continue returning y_0 . Their evidence reliance differs although their single-world observation is identical. Therefore reliance is not identifiable from that observation alone. $\square$

The proposition does not say observational metrics are unhelpful. It says they identify a different property. Entailment protects the reader against unsupported claims in the displayed artifact; intervention tests whether the system behavior is governed by the evidence. A trustworthy evidence-using system needs both.

4 The Causal Evidence Audit

4.1 Matched evidence worlds

For each base item, an audit author specifies a query q , two evidence worlds $E^{(0)}$ and $E^{(1)}$, distinct oracle answers $y^{(0)}$ and $y^{(1)}$, and oracle proof sets $G^{(0)}$ and $G^{(1)}$. The worlds differ by a minimal, answer-determining edit T such that

$$E^{(1)} = T(E^{(0)}), \quad y^{(1)} \neq y^{(0)}. \quad (2)$$

All non-causal features that can feasibly be controlled—document count, style, entity frequency, answer mention frequency, and ordering—should remain fixed or be explicitly balanced by an answer-invariant edit whose role is documented.

The audit also constructs an ablated world $E^{(-)}$ by removing an indispensable proof element from $E^{(0)}$. The oracle answer is then $\perp$. An optional nuisance intervention P permutes irrelevant documents or changes non-semantic metadata while preserving the oracle answer. This fourth condition tests selective invariance, but it was not included in the pilot.

4.2 Metrics

Let $(\hat{y}^{(w)}, C^{(w)}) = A(q, E^{(w)})$. Exact answer matching can be replaced by a task-appropriate semantic judge, provided the judge is fixed before evaluation.

Counterfactual response consistency.

$$\text{CRC}_i = \mathbb{I}[\hat{y}_i^{(0)} = y_i^{(0)} \wedge \hat{y}_i^{(1)} = y_i^{(1)}]. \quad (3)$$

Because the two oracle answers differ, this tests a correct answer switch rather than mere output instability.

Evidence-necessity abstention.

$$\text{ENA}_i = \mathbb{I}[\hat{y}_i^{(-)} = \perp]. \quad (4)$$

ENA penalizes filling an evidence gap with an unsupported answer. It is not a generic refusal metric: abstention is correct only in the matched ablation world.

Proof citation precision and recall. For $w \in \{0, 1\}$,

$$\text{PCR}_i^{(w)} = \frac{|C_i^{(w)} \cap G_i^{(w)}|}{|G_i^{(w)}|}, \quad (5)$$

$$\text{PCP}_i^{(w)} = \frac{|C_i^{(w)} \cap G_i^{(w)}|}{|C_i^{(w)}|}, \quad (6)$$

with PCP defined as zero for an empty citation set when G is nonempty. PCR asks whether every necessary proof document is cited; PCP penalizes extraneous citations. Let $V_i^{(-)}$ indicate that every citation emitted in the ablated world names a document actually present in that world. Span-level proof sets should be preferred when reliable annotations are available.

Joint scores. The coverage-based score is a hard per-item pass:

$$\text{CES}_i^{\text{cov}} = \text{CRC}_i \text{ENA}_i V_i^{(-)} \prod_{w \in \{0, 1\}} \mathbb{I}[\text{PCR}_i^{(w)} = 1]. \quad (7)$$

The strict score additionally requires exact proof citations in both full worlds and no proof citation after abstention:

$$\begin{aligned} \text{CES}_i^{\text{strict}} &= \text{CRC}_i \text{ENA}_i V_i^{(-)} \mathbb{I}[C_i^{(-)} = \emptyset] \\ &\times \prod_{w \in \{0, 1\}} \mathbb{I}[C_i^{(w)} = G_i^{(w)}]. \end{aligned} \quad (8)$$

Dataset scores are macro-averages over items. The multiplicative construction is intentional: high answer accuracy cannot compensate for guessing when evidence is absent, and citation verbosity cannot compensate for omitting a necessary proof link.

Single-world observational pass. For comparison, the pilot reports

$$\text{OBS}_i = \mathbb{I}[\hat{y}_i^{(0)} = y_i^{(0)}] \mathbb{I}[\text{PCR}_i^{(0)} = 1]. \quad (9)$$

OBS is a controlled proxy for the common pattern “correct answer plus complete supporting citations.” It is not claimed to reproduce every existing attribution metric.

4.3 Audit validity requirements

An intervention is only informative if the counterfactual world is coherent and the edit is truly answer-determining. We recommend five construction gates:

1. **Minimality:** change the fewest atomic propositions consistent with a coherent alternative world.
2. **Answer separation:** independently verify that the oracle answers differ.
3. **Proof completeness:** annotate all necessary sources, including multi-hop links.
4. **Nuisance balance:** control position, length, lexical overlap, source style, and answer-mention frequency.
5. **Ablation validity:** confirm that removal makes the answer underdetermined rather than merely harder.

For open-web agents, exact control is difficult. A practical alternative is to freeze a retrieved snapshot, perform document-level or claim-level interventions inside the snapshot, and evaluate the synthesis stage separately from retrieval. End-to-end retrieval interventions can then be added as a second layer.

5 Pilot Study

5.1 Research questions

The pilot addresses three narrow questions:

- RQ1.** Does paired intervention expose a gap hidden by a single-world observational pass?
- RQ2.** Is missing-evidence abstention a distinct bottleneck from following a changed fact?
- RQ3.** Does a stronger evidence-contract prompt reliably improve the joint score?

5.2 Dataset

CEVAL-32 contains 32 items, eight in each of four reasoning families:

- **Direct lookup:** one registry sentence determines a fictional reactor’s authorized coolant.
- **Two-hop composition:** a dispatch ledger selects a depot, and a second source maps the depot to a cipher.
- **Numerical comparison:** two verified values and a ranking rule determine the higher-ranked fictional station.
- **Conditional reasoning:** a threshold policy and a site measurement determine a required protocol.

All entity and value names were constructed for the benchmark. This reduces the opportunity to answer from memorized world knowledge. For each item, world 1 changes one decisive sentence or number from world 0. In the direct, two-hop, and conditional families, all other documents are unchanged. In the comparison family, the

explicitly obsolete value in D4 also swaps high and low tokens to preserve the numerical multiset across worlds; D4 is excluded from the oracle proof and does not affect the answer under the benchmark semantics. The ablation removes the decisive document from world 0. Distractors include obsolete estimates, preliminary trials, availability statements, and non-diagnostic descriptions. The benchmark file records both worlds, expected answers, gold citation sets, and the ablation target.

5.3 Models and prompting regimes

We evaluated the 4-bit MLX conversions `mlx-community/Qwen3-4B-4bit` and `mlx-community/Llama-3.2-3B-Instruct-4bit`. Qwen3 is documented in Yang et al. [2025], while the exact Llama-3.2-3B-Instruct release is documented by its model card [Meta, 2024]. The two evaluated MLX conversion artifacts are identified separately [MLX Community, 2025, 2024], and their immutable revisions are recorded in the public repository. Quantized conversions may not reproduce full-precision behavior, so these identifiers are part of the experimental specification rather than incidental details.

Each model was tested under two system prompts. The *baseline* required use of numbered sources, abstention when sources were insufficient, and a JSON answer with citation IDs. The *evidence contract* additionally required the model to identify internally the smallest complete proof chain, answer only when every necessary link was explicit, and cite every proof source without irrelevant citations. Full prompts appear in Appendix A.

The design comprises $32 \times 3 \times 2 \times 2 = 384$ generations: 32 items, three evidence conditions, two models, and two regimes. Decoding was greedy (temperature zero) with a 96-token maximum. All inference was local through MLX; no external API was used. The release run used Python 3.14.6 on a 14-core Apple M4 Pro with 48 GB memory, running macOS 26.5.2, MLX 0.32.2, MLX-LM 0.31.3, and Transformers 5.17.0. The experiment script writes the generated benchmark, per-model raw responses, summary metrics, and a machine-readable provenance manifest with source and artifact hashes.

Code and artifacts. The complete source code for the tests is available at <https://github.com/sergiofigueras/causal-evidence-audit>; the immutable artifact release is <https://github.com/sergiofigueras/causal-evidence-audit/releases/tag/v0.1.0>. The repository contains the benchmark generator, the realized 32-item benchmark, both system prompts, all 384 raw generations, machine-readable summaries, model identifiers and revisions, a hash-locked environment, validation code, and execution instructions.

5.4 Scoring and uncertainty

Answers were normalized for case and punctuation. A response counted as correct if it contained exactly the expected fictional candidate and not the alternative; wrapper phrases such as “Protocol X” were allowed. Abstentions were recognized from a fixed list including `INSUFFICIENT`, “not specified,” and “cannot determine.” Citation IDs were validated against the JSON schema. Gold proof coverage required every annotated proof source; strict matching rejected both missing and extraneous sources.

We report Wilson 95% confidence intervals for binary proportions. These intervals quantify sampling uncertainty over the 32 templated items, not uncertainty over prompts, decoders, model versions, domains, or natural document distributions. No null-hypothesis significance tests are reported because the pilot was designed for diagnostic demonstration, not comparative model inference.

6 Results

6.1 RQ1: observational success overstates joint grounding

Across the four system configurations, OBS ranged from 59.4% to 78.1%, while CES^{cov} ranged from 0% to 31.3% (Table 1). For Qwen3-4B with the baseline prompt, OBS was 75.0% (Wilson 95% CI 57.9–86.7), but the coverage-based joint score was 31.3% (18.0–48.6). For Llama-3.2-3B with the baseline prompt, OBS was 59.4% (42.3–74.5) and the joint score was 0% (0–10.7).

This gap is the primary pilot finding. In a conventional single-world inspection, many responses had the expected answer and cited every gold proof source. The matched ablation revealed that the same system frequently answered when the evidence did not determine an answer. Thus, the issue was not simply failure to read a changed fact: CRC was 87.5% for both baseline systems. The audit separates evidence following from evidence necessity.

6.2 RQ2: abstention is the limiting component

ENA ranged from 0% to 31.3%, substantially below paired-world answer accuracy in every configuration. The Qwen baseline abstained correctly on 10 of 32 ablations. The Llama baseline abstained on none, despite answering both full evidence worlds correctly on 28 of 32 pairs. These are classic unsupported-completion cases: correct behavior while evidence is present coexists with guessing when a proof link is absent.

Family-level strict scores sharpen the diagnosis. Under the Qwen baseline, direct lookup passed on all 8 items, conditional reasoning passed on 2 of 8, and both two-hop and comparison families passed on 0 of 8. The ablated

Table 1: Pilot results on 32 items per model and prompt regime. W0/W1 are answer accuracy in the paired worlds; CRC is paired response consistency; ENA is evidence-necessity abstention; OBS is the single-world observational pass; CES^{cov} requires correct switching, valid ablation citations, ablation abstention, and complete proof citations in both worlds; CES^{strict} also rejects extraneous full-world citations and any ablation citation. Values are percentages.

Model	Prompt	W0	W1	CRC	ENA	OBS	CES ^{cov}	CES ^{strict}
Qwen3-4B	Baseline	87.5	100.0	87.5	31.3	75.0	31.3	31.3
Qwen3-4B	Evidence contract	87.5	100.0	87.5	6.3	78.1	6.3	0.0
Llama-3.2-3B	Baseline	93.8	93.8	87.5	0.0	59.4	0.0	0.0
Llama-3.2-3B	Evidence contract	90.6	90.6	81.3	21.9	71.9	6.3	0.0

two-hop and comparison contexts retained plausible candidate mentions, which appear to invite completion from partial evidence. This pattern should be treated as a hypothesis for a larger study, because templates within a family share structure.

6.3 RQ3: stronger instructions do not guarantee causal grounding

The evidence-contract prompt increased OBS by 3.1 percentage points for Qwen and 12.5 points for Llama, but it did not reliably increase the joint scores. For Qwen, ENA fell from 31.3% to 6.3%, CES^{cov} fell by the same amount, and CES^{strict} fell to zero. For Llama, ENA improved from 0% to 21.9% and CES^{cov} reached 6.3%, yet the strict score remained zero.

The coverage-strict gap localizes over-citation. Two Qwen evidence-contract items and two Llama evidence-contract items satisfied correct switching, valid citations under ablation, correct ablation abstention, and complete gold coverage; none satisfied the stricter citation contract across all three conditions. The evidence contract often elicited additional, non-necessary sources. An evaluation based only on whether all required citations were present would reward this verbosity, while strict proof alignment exposes it.

These prompt comparisons are exploratory. Greedy decoding, one prompt wording, small models, and 32 templates cannot establish a general effect of evidence contracts. They do demonstrate a practical reason to evaluate the target property directly: a prompt can improve the visible citation artifact without improving the intended evidence behavior.

7 From Evaluation to Training

The paired construction naturally yields a training objective. Let $p_\theta(y \mid q, E)$ be the answer distribution, $p_\theta(c_j = 1 \mid q, E, y)$ the probability of citing source j , and $p_\theta^E = p_\theta(\cdot \mid q, E)$. A causal-grounding objective can

augment the usual task loss:

$$\begin{aligned} \mathcal{L}_{\text{CEA}} = \mathcal{L}_{\text{task}} &- \lambda_{\text{cf}} \sum_{w \in \{0,1\}} \log p_\theta(y^{(w)} \mid q, E^{(w)}) \\ &- \lambda_{\text{abs}} \log p_\theta(\perp \mid q, E^{(-)}) \\ &+ \lambda_{\text{inv}} D_{\text{KL}}(p_\theta^E \parallel p_\theta^{P(E)}) \\ &+ \lambda_{\text{cite}} \sum_j \text{BCE}(p_\theta(c_j = 1), \mathbb{I}[j \in G]). \quad (10) \end{aligned}$$

The counterfactual term teaches correct response to decisive changes; the abstention term teaches necessity; the optional invariance term discourages sensitivity to nuisance permutations; and the citation term targets the proof set rather than citation volume.

The same data can support preference optimization without requiring access to logits. For each query, a preferred trajectory answers both coherent worlds correctly, abstains under ablation, and cites the minimal proof. A dispreferred trajectory may keep the original answer after a fact flip, guess after ablation, omit a proof link, or pad citations. Importantly, training and evaluation interventions should be generated from disjoint templates and domains to prevent learning a superficial “counterfactual benchmark” pattern.

For long-horizon research agents, interventions can be applied at several resolutions:

- **Claim level:** alter a decisive number, date, entity relation, or experimental outcome in a frozen source snapshot.
- **Document level:** replace a source with a matched alternative and update downstream oracle conclusions.
- **Dependency level:** intervene on an intermediate node in an annotated research graph and test all affected conclusions.
- **Retrieval level:** alter corpus availability or ranking while checking whether the agent recovers equivalent evidence elsewhere.

Mr.LHDR’s dependency graphs [Guo et al., 2026] are especially compatible with the third design. ReportBench and DeepResearch Bench could add paired source snapshots to distinguish reports that merely contain support from reports whose conclusions track the supplied literature [Li et al., 2025, Du et al., 2025].

8 Discussion

8.1 What the audit establishes

Passing CEA provides behavioral evidence that, within a controlled item family, the system tracks answer-determining evidence, recognizes when a required link is absent, and exposes the proof sources it used. This is stronger than observing an entailing citation once. It is still a behavioral criterion, not a mechanistic causal explanation of internal neural computation. The word “causal” refers to interventions on the system’s evidence input and their effect on its output under a specified benchmark causal model.

8.2 What the audit does not establish

First, a model can faithfully follow false or malicious evidence. Source truth, authority, independence, and timelessness require separate evaluation. GopherCite already noted that evidential support is only one part of trustworthiness [Menick et al., 2022], while work on knowledge conflicts and poisoning studies what happens when sources disagree or are adversarial [Wang et al., 2023, Kumar et al., 2026]. Second, passing controlled items does not imply robust open-web research. Retrieval may fail before synthesis begins. Third, citation necessity is task-relative: a proof set minimal under one annotation scheme may not be the only valid proof.

8.3 Why a hard joint score

Composite averages can hide catastrophic modes. A system with excellent full-context accuracy but no abstention can appear strong when metrics are averaged. CES uses a conjunction because high-stakes evidence use is a contract: the system should follow decisive evidence, avoid claims when the proof is incomplete, and reveal the proof it relied on. Component metrics should always be reported beside the joint score so developers can diagnose which part failed.

9 Limitations and Threats to Validity

Pilot scale and templating. The dataset has 32 English items generated from four templates. Items within a family are not independent draws from natural research tasks. Wilson intervals therefore describe item-level variation only and may be over-optimistic about domain generalization.

Model coverage. Only two small, 4-bit open-weight models were tested. Quantization, chat templates, and MLX implementation can affect behavior. Results must not be extrapolated to larger open models, proprietary models, or complete deep-research products.

No retrieval stage. The pilot freezes the evidence bundle and evaluates synthesis. This is deliberate for intervention control, but it excludes query formation, source discovery, ranking, browsing, and source-quality judgments. A complete agent audit should report retrieval and synthesis layers separately and jointly.

Ablation artifacts. Removing a document changes context length and positions. A model might respond to these surface changes rather than evidence semantics. Future work should use matched placeholders, position-balanced removals, and irrelevant-document replacements, then report invariance under permutations.

Oracle proof sets. The pilot’s proofs are unambiguous by construction, but real documents admit alternative derivations and source redundancy. Real-world extensions need multiple valid proof sets, span-level annotations, and adjudication by domain experts.

Answer and abstention parsing. The deterministic parser accepts fixed linguistic variants. It may misclassify unusual but reasonable responses. Human validation and preregistered semantic judges are needed for open-ended reports.

Execution stability. Temperature-zero greedy decoding removes sampling randomness but does not imply byte-level determinism across software, kernels, chat templates, or hardware. The release preserves raw outputs and a provenance manifest, and the comparison command distinguishes aggregate-metric reproduction from exact raw-text identity.

Novelty boundary. The protocol synthesizes ideas from citation evaluation, context attribution, counterfactual risk control, knowledge-conflict evaluation, and dependency-aware research benchmarks. Its contribution is the joint end-to-end contract and matched evaluation design, not the invention of input ablation or counterfactual testing in isolation.

10 Research Integrity and Ethical Use

All numerical claims in this manuscript are generated by the included script and preserved raw outputs. No human-subject data, private data, or external API calls were used. Fictional entities reduce privacy and memorization concerns. The paper and literature survey were prepared with AI assistance; responsibility for the claims, citations, released code, and artifacts remains with the named author. This preprint has not been peer reviewed and should not be represented as accepted or independently validated research.

The method can improve safety by exposing unsupported completion, but it can also be misused to optimize systems narrowly against known intervention templates. Public benchmarks should therefore publish transparent development sets while maintaining structurally distinct held-out audits and periodic refreshes.

11 Conclusion

A citation can make an answer verifiable without proving that the answer was grounded in the cited source. That distinction matters as RAG systems and research agents produce increasingly polished, citation-rich outputs. CEA adds a simple requirement: change the decisive evidence and the answer should change; remove indispensable evidence and the system should abstain; inspect the citations and they should match the proof.

The CEVAL-32 pilot shows why this layer is useful. Configurations that passed a single-world answer-and-citation check on roughly three-fifths to four-fifths of items achieved much lower joint causal scores, largely because they guessed when evidence was incomplete. Stronger instructions did not reliably repair the behavior. Larger, expert-validated studies are required, but the evaluation principle is immediately actionable: claims of grounded generation should be supported by paired evidence interventions, not citation entailment alone.

Data and Code Availability

The source code for the tests and the complete reproducibility artifact are publicly available at <https://github.com/sergiofigueras/causal-evidence-audit>. Version 0.1.0 is preserved at <https://github.com/sergiofigueras/causal-evidence-audit/releases/tag/v0.1.0>. The repository contains the benchmark generator, the realized 32-item benchmark, all 384 raw generations, machine-readable summaries, model identifiers and revisions, prompts, a provenance manifest, a hash-locked environment, validation code, and execution instructions. The artifact is designed to run locally on Apple Silicon through MLX and is released under the MIT License.

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A Prompt Templates

Baseline system prompt.

Answer the question using only the numbered sources. If the sources do not determine the answer, answer INSUFFICIENT. Return exactly one

JSON object with keys answer and citations. citations must be a list of source IDs such as ["D1"]. Do not add prose.

Evidence-contract system prompt.

Use only the numbered sources. Internally identify the smallest complete proof chain for the answer. Answer only when every necessary link is explicitly supported; otherwise answer INSUFFICIENT. Treat the current sources as authoritative even when their entities are unfamiliar. Return exactly one JSON object with keys answer and citations. citations must list every source in the proof chain and no irrelevant source. Do not add prose.

User prompt schema.

```
SOURCES
[D1] <document text>
...
[Dm] <document text>

QUESTION
<query>
```

B Worked Audit Example

Consider a fictional two-hop item. In world 0, D1 routes courier Vela through Aldermere, D2 states that Aldermere uses the Galdite cipher, and D3 states that Bramblecross uses Hesper. The correct answer is Galdite with proof set {D1,D2}. In world 1, only D1 changes and routes Vela through Bramblecross; the answer must change to Hesper with proof set {D1,D3}. In the ablation, D1 is removed, so the depot is unknown and the correct output is INSUFFICIENT. A response that answers Galdite in world 0 with D1 and D2 passes the observational check. It passes the causal audit only if it also answers Hesper with D1 and D3 in world 1 and abstains under ablation.

C Uncertainty Intervals

Table 2: Wilson 95% intervals for selected pilot metrics; $n = 32$ per row.

Model	Prompt	OBS	CES ^{cov}
Qwen3	Base	57.9–86.7	18.0–48.6
Qwen3	Contract	61.2–89.0	1.7–20.2
Llama	Base	42.3–74.5	0.0–10.7
Llama	Contract	54.6–84.4	1.7–20.2

D Recommended Reporting Checklist

An evaluation using CEA should report:

1. the intervention unit and minimality rule;
2. how coherence and oracle answer separation were verified;
3. all valid proof sets and citation granularity;
4. position, length, style, and lexical controls;
5. model version, retrieval snapshot, prompt, decoder, and tool configuration;
6. CRC, ENA, citation precision/recall, CES^{cov} , and CES^{strict} separately;
7. uncertainty intervals and the sampling unit they actually describe;
8. raw outputs, parser or judge code, and adjudication rules; and
9. limitations covering source truth, retrieval failure, and external validity.

References

- Amin Abolghasemi, Leif Azzopardi, Seyyed Hadi Hashemi, Maarten de Rijke, and Suzan Verberne. Evaluation of attribution bias in generator-aware retrieval-augmented large language models. *arXiv preprint arXiv:2410.12380*, 2024. URL <https://arxiv.org/abs/2410.12380>.
- Cedric Caruzzo, Donggeun Yoo, and Tae Soo Kim. Deceptive grounding: Entity attribution failure in clinical retrieval-augmented generation. *arXiv preprint arXiv:2607.09349*, 2026. URL <https://arxiv.org/abs/2607.09349>.
- Lu Chen, Ruqing Zhang, Jiafeng Guo, Yixing Fan, and Xueqi Cheng. Controlling risk of retrieval-augmented generation: A counterfactual prompting framework. *arXiv preprint arXiv:2409.16146*, 2024. URL <https://arxiv.org/abs/2409.16146>.
- Benjamin Cohen-Wang, Harshay Shah, Kristian Georgiev, and Aleksander Madry. ContextCite: Attributing model generation to context. *arXiv preprint arXiv:2409.00729*, 2024. URL <https://arxiv.org/abs/2409.00729>.
- Mingxuan Du, Benfeng Xu, Chiwei Zhu, Xiaorui Wang, and Zhen-dong Mao. Deepresearch bench: A comprehensive benchmark for deep research agents. *arXiv preprint arXiv:2506.11763*, 2025. URL <https://arxiv.org/abs/2506.11763>.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023. URL <https://arxiv.org/abs/2309.15217>.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. *arXiv preprint arXiv:2305.14627*, 2023. URL <https://arxiv.org/abs/2305.14627>.
- Minghao Guo, Meng Cao, Sui Zhao, Siyu Ning, Xin Wang, Haoze Zhao, Jiaxuan Yang, Haihong Hao, Mingfei Han, Shunlin Rong, et al. Mr.LHDR: A benchmark for multimodal real-world long-horizon deep research agents. *arXiv preprint arXiv:2609.11318*, 2026. URL <https://arxiv.org/abs/2609.11318>.
- Wenyue Hua, Jiang Guo, Mingwen Dong, Henghui Zhu, Patrick Ng, and Zhiguo Wang. Propagation and pitfalls: Reasoning-based assessment of knowledge editing through counterfactual tasks. *arXiv preprint arXiv:2401.17585*, 2024. URL <https://arxiv.org/abs/2401.17585>.
- Pushkal Kumar, Tucker Nielson, Tanish Kolhe, Shubham Zala, and Vincent Li. RAGuard: A layered defense framework for retrieval-augmented generation systems against data poisoning. *arXiv preprint arXiv:2607.26339*, 2026. URL <https://arxiv.org/abs/2607.26339>.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2005.11401>.
- Minghao Li, Ying Zeng, Zhihao Cheng, Cong Ma, and Kai Jia. Report-Bench: Evaluating deep research agents via academic survey tasks. *arXiv preprint arXiv:2508.15804*, 2025. URL <https://arxiv.org/abs/2508.15804>.
- Nelson F. Liu, Tianyi Zhang, and Percy Liang. Evaluating verifiability in generative search engines. *arXiv preprint arXiv:2304.09848*, 2023. URL <https://arxiv.org/abs/2304.09848>.
- Jacob Menick, Maja Trebacz, Vladimir Mikulik, John Aslanides, Francis Song, Martin Chadwick, Mia Glaese, Susannah Young, Lucy Campbell-Gillingham, Geoffrey Irving, and Nat McAleese. Teaching language models to support answers with verified quotes. *arXiv preprint arXiv:2203.11147*, 2022. URL <https://arxiv.org/abs/2203.11147>.
- Meta. Llama 3.2: 3B instruct model card. Hugging Face model card, 2024. URL <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>. Accessed September 12, 2026.
- MLX Community. Llama-3.2-3B-Instruct-4bit model card. Hugging Face model repository, 2024. URL <https://huggingface.co/mlx-community/Llama-3.2-3B-Instruct-4bit>. Accessed September 12, 2026.
- MLX Community. Qwen3-4B-4bit model card. Hugging Face model repository, 2025. URL <https://huggingface.co/mlx-community/Qwen3-4B-4bit>. Accessed September 12, 2026.
- Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, et al. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*, 2021. URL <https://arxiv.org/abs/2112.09332>.
- Jingxi Qiu, Zeyu Han, and Cheng Huang. SURE-RAG: Sufficiency and uncertainty-aware evidence verification for selective retrieval-augmented generation. *arXiv preprint arXiv:2605.03534*, 2026. URL <https://arxiv.org/abs/2605.03534>.
- Hannah Rashkin, Vitaly Nikolaev, Matthew Lamm, Lora Aroyo, Michael Collins, Dipanjan Das, Slav Petrov, Gaurav Singh Tomar, Iulia Turc, and David Reitter. Measuring attribution in natural language generation models. *arXiv preprint arXiv:2112.12870*, 2021. URL <https://arxiv.org/abs/2112.12870>.
- Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, Jiayang Cheng, Cunxiang Wang, Shichao Sun, Huanyu Li, et al. RAGCHECKER: A fine-grained framework for diagnosing retrieval-augmented generation. *arXiv preprint arXiv:2408.08067*, 2024. URL <https://arxiv.org/abs/2408.08067>.
- Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. *arXiv preprint arXiv:2311.09476*, 2023. URL <https://arxiv.org/abs/2311.09476>.
- Rishiraj Saha Roy, Joel Schlotthauer, Chris Hinze, Andreas Foltyn, Luzian Hahn, and Fabian Kuech. Evidence contextualization and counterfactual attribution for conversational QA over heterogeneous data with RAG systems. *arXiv preprint arXiv:2412.10571*, 2024. URL <https://arxiv.org/abs/2412.10571>.
- Yash Saxena, Raviteja Bommireddy, Ankur Padia, and Manas Gaur. Generation-time vs. post-hoc citation: A holistic evaluation of LLM attribution. *arXiv preprint arXiv:2509.21557*, 2025. URL <https://arxiv.org/abs/2509.21557>.
- Yike Wang, Shangbin Feng, Heng Wang, Weijia Shi, Vidhisha Balachandran, Tianxing He, and Yulia Tsvetkov. Resolving knowledge conflicts in large language models. *arXiv preprint arXiv:2310.00935*, 2023. URL <https://arxiv.org/abs/2310.00935>.
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for LLMs: A survey. *arXiv preprint arXiv:2403.08319*, 2024. URL <https://arxiv.org/abs/2403.08319>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Xinye Yang, Zhenyang Liu, Ruisi Li, and Yuanyuan Lei. Large language models in resolving contextual knowledge conflicts. *arXiv preprint arXiv:2609.03148*, 2026. URL <https://arxiv.org/abs/2609.03148>.
- Dingling Zhang, He Zhu, Jincheng Ren, Kangqi Song, Xinran Zhou, Boyu Feng, Shudong Liu, Jiabin Luo, Weihao Xie, Zhaohui Wang, et al. How far are we from genuinely useful deep research agents? *arXiv preprint arXiv:2512.01948*, 2025. URL <https://arxiv.org/abs/2512.01948>.
- Zexuan Zhong, Zhengxuan Wu, Christopher D. Manning, Christopher Potts, and Danqi Chen. MQuAKE: Assessing knowledge editing in language models via multi-hop questions. *arXiv preprint arXiv:2305.14795*, 2023. URL <https://arxiv.org/abs/2305.14795>.